

Project 2 Report: Computer Simulation - Dr. Ghatari

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1. Introduction and Problem Statement

Dataset and Target Variable Introduction: To conduct this project, the “**Water Potability**” dataset was selected. Among the available features, the continuous variable **pH** was chosen as the target variable for modeling. Exiting the standard range for this variable represents a dangerous event, making the water undrinkable.

Exact Determination of the Rare Event Threshold: In order to have a scientific definition consistent with the real behavior of the data, the critical threshold was defined as falling in the **first 1%** and the **last 1%** of the distribution. By calculating the percentiles on the raw data, the threshold values were extracted as follows:

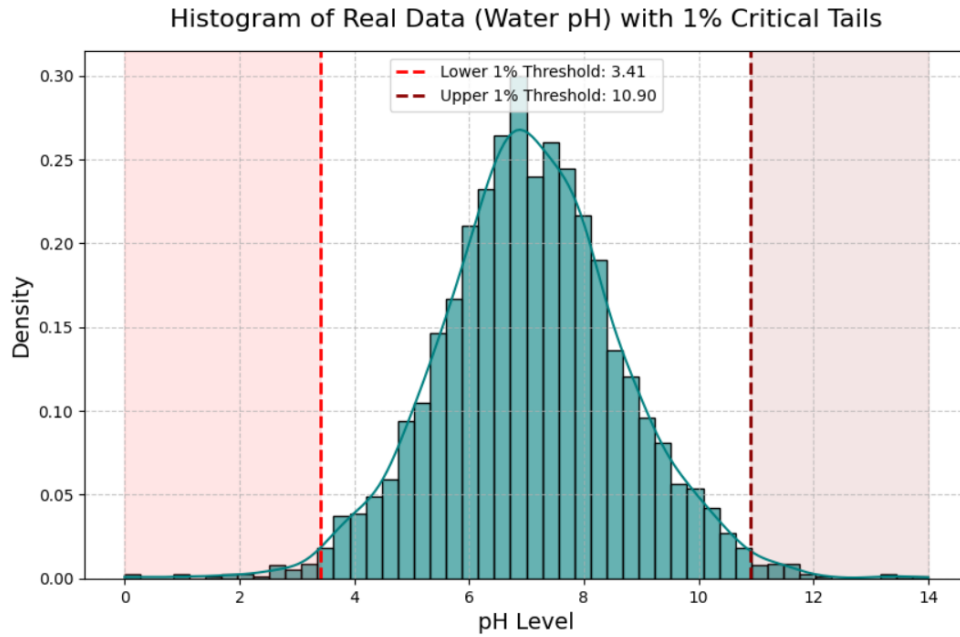
- **Lower Threshold (Acidity Crisis):** pH value less than 4.34
- **Upper Threshold (Alkalinity Crisis):** pH value greater than 9.75

Based on the above definition, the total probability of the rare event occurring in the initial raw data is equal to **2 percent**.

2. Statistical Analysis and Distribution Fitting

To model the pH variable and simulate values outside the range of available samples, it is necessary to find a continuous Probability Density Function (PDF), $f(y)$, that best describes the behavior of the empirical data. The analysis steps are as follows:

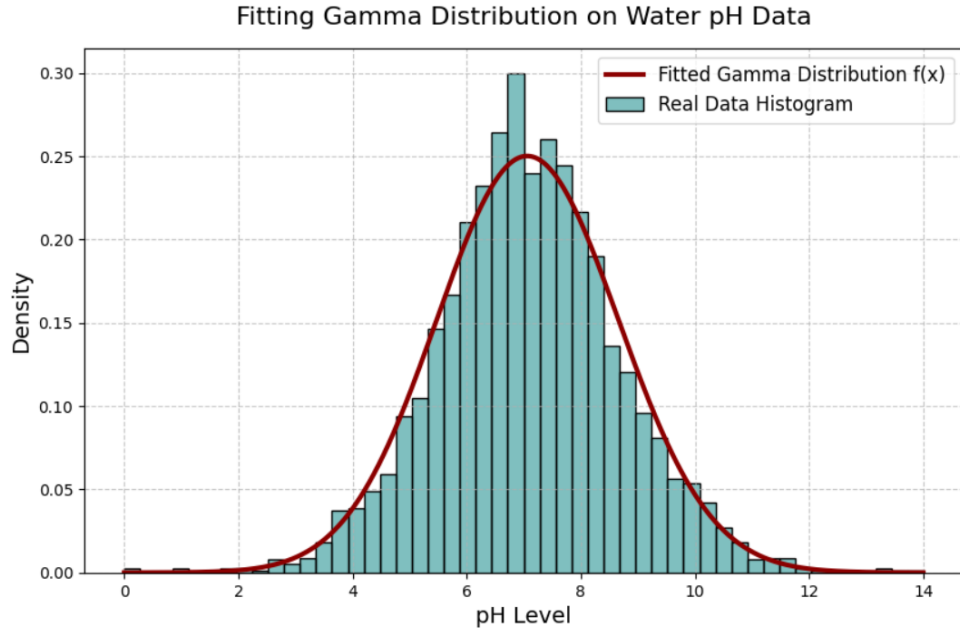
Plotting the Histogram of Real Data: First, data cleaning was performed, and the density histogram of the real data was plotted. Visual inspection of the histogram shows that the data distribution is unimodal and bell-shaped, but not strictly perfectly aligned with a standard normal distribution.



Selection and Fitting of the Distribution: Considering the continuous and positive nature of pH values and the true data distribution, the **Gamma distribution** was selected for fitting. Using Python's statistical functions, the optimal parameters for this distribution were calculated over the data:

```
--- Gamma Distribution Parameters ---  
Shape (a): 6615.4484  
Loc (Shift): -122.5883  
Scale (Scale): 0.0196
```

The extremely large alpha (Shape) parameter indicates that the behavior of the fitted Gamma distribution has become highly similar to a Normal distribution. Furthermore, the mean of this distribution is evaluated at 7.07, which completely matches the mean of the real data.



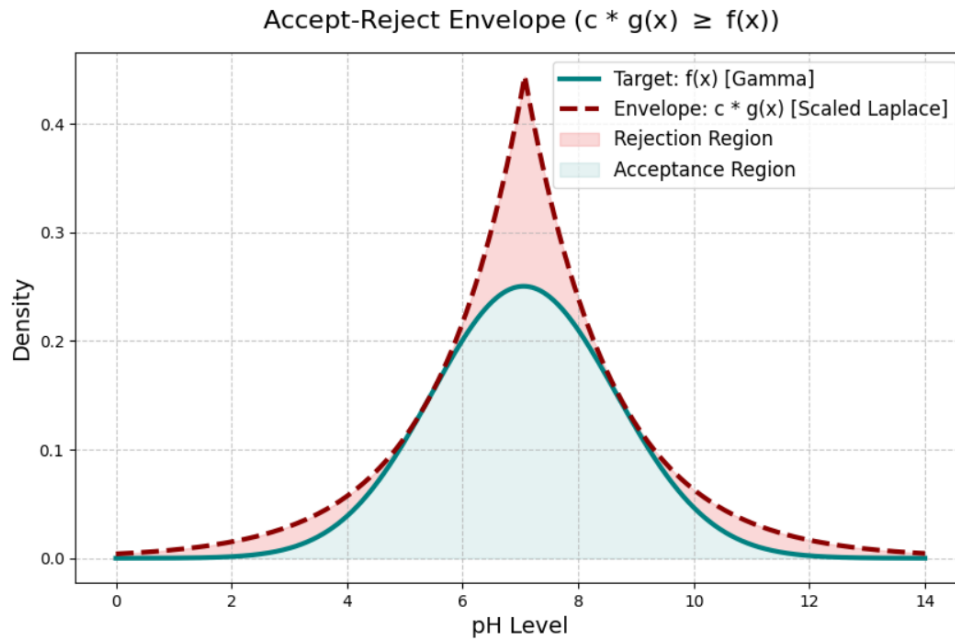
3. Accept-Reject Algorithm Implementation

To generate data, the Accept-Reject algorithm was utilized.

Introducing the Proposal Distribution ($g(y)$): According to the course notes, for covering bell-shaped distributions, using the Laplace distribution is more appropriate due to its heavier and exponential tails. Since the fitted Gamma distribution in this project behaves very similarly to a Normal distribution, I chose the Laplace distribution as the proposal candidate to ensure it has heavier tails relative to the target distribution, keeping the fraction $\frac{f(y)}{g(y)}$ bounded.

The center of this distribution was set on the mean of the Gamma distribution (7.07), and the scale parameter was set to 1.5. Sample generation from the Laplace distribution was implemented in Python code via the Inverse Transform method.

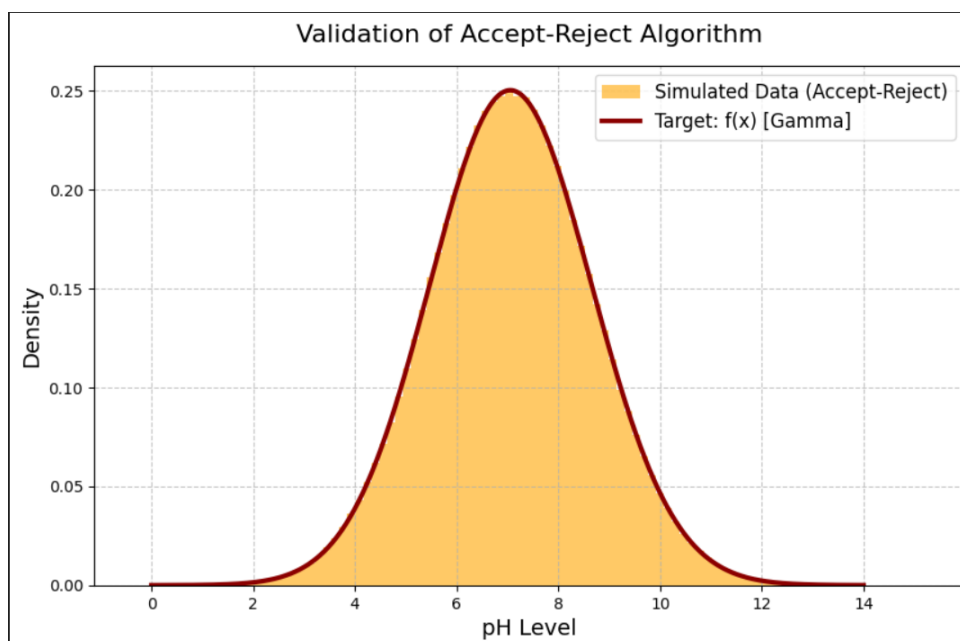
Calculating the Scaling Factor (c): The optimal value for factor c was obtained by calculating the maximum ratio of the two functions, i.e., $c = \sup \frac{f(y)}{g(y)}$. Based on the algorithm's calculations, this value equals **1.3314**.



Checking the Acceptance Rate: The theoretical acceptance probability is $p = \frac{1}{c} = \frac{1}{1.3314} \approx 0.7511$ or **75%**.

In the practical implementation, to generate 1,000,000 data points, the `while` loop was executed 1,331,073 times, demonstrating that the practical acceptance rate matches its theoretical counterpart perfectly.

```
Total samples needed to generate 1000000 valid points: 1331073
Theoretical Acceptance Rate (1/c): 0.7511 (75.11%)
Actual Acceptance Rate: 0.7513 (75.13%)
```



Algorithm Efficiency Analysis: To analyze the efficiency of the Accept-Reject algorithm, we must consider the behavior of the proposal distribution (Laplace) relative to the target distribution (Gamma) and the scaling constant c . The acceptance probability of a sample in this algorithm is $p = \frac{1}{c}$.

The obtained value of $p = 0.75$ means that, on average, out of 4 random data points generated, 3 are accepted and only 1 is rejected. This was also proven in practice, where the algorithm ran 1,331,073 times for one million valid data points. If the candidate distribution had lighter tails, the value of c would increase, leading to lower efficiency.

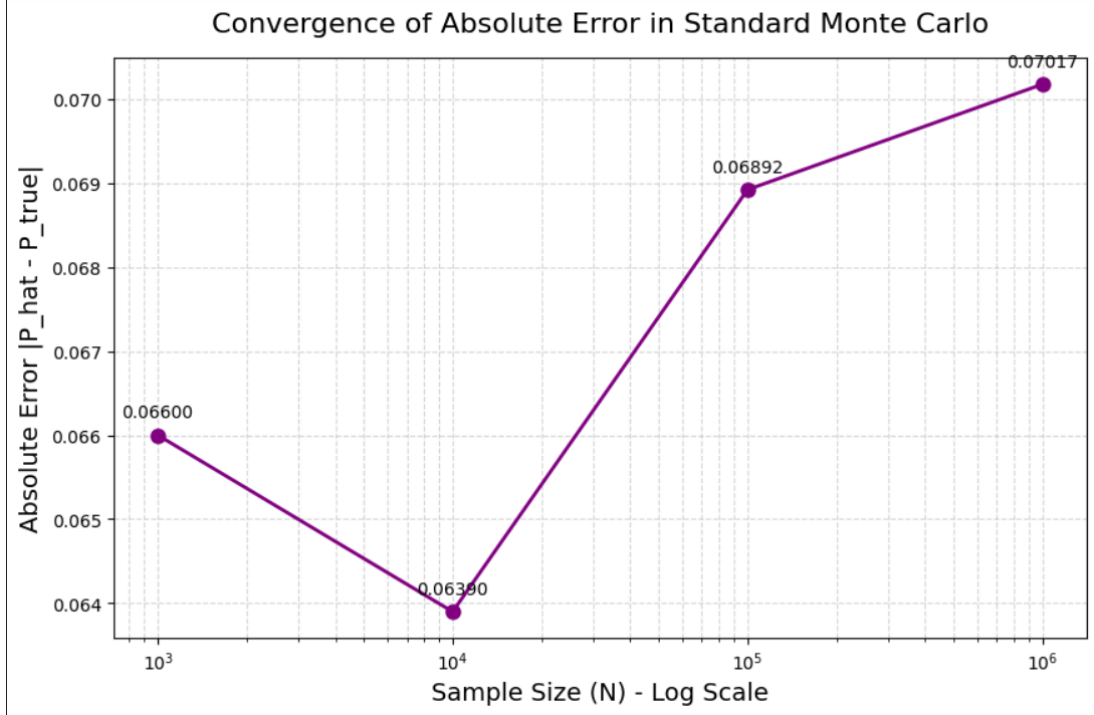
4. Standard Monte Carlo Simulation

In this section, the one million simulated data points generated in the previous step by the Accept-Reject algorithm were utilized.

Executing Simulation and Calculating Probability: The probability calculation is equivalent to calculating the expected value of an indicator function; such that if sample y_i falls in the critical region, the function's value is 1, and otherwise, it is 0.

The estimated probability was calculated by averaging these values. The simulation was run for four distinct sample sizes. The results showed that as N increases, the estimated value converges towards the theoretical value of the Gamma distribution (approx. 0.047), and the standard error continuously decreases.

--- Standard Monte Carlo Results ---					
N (Sample Size)	Estimated Prob (P_hat)	Absolute Error	MC Variance (S^2/N)	Number of Rare Events	
10^3	0.086000	0.066000	7.860400e-05	86	
10^4	0.083900	0.063900	7.686079e-06	839	
10^5	0.088920	0.068920	8.101323e-07	8892	
10^6	0.090174	0.070174	8.204265e-08	90174	



Investigating High Variance and Inefficiency for Rare Events: Despite the algorithm's convergence with increasing sample size, the Standard Monte Carlo method is highly inefficient for estimating rare events. The reason is that since the targeted event is extremely rare, the vast majority of generated samples fall in the safe region, yielding a zero value for the indicator function. This means a significant amount of computational time is wasted on processing data that plays no role in estimating the critical event.

Furthermore, for rare events, the simulation variance causes the relative error to become exceptionally large. Consequently, to achieve a good approximation, the SMC algorithm requires the generation of billions of samples, which is neither logical nor cost-effective.

5. Importance Sampling

Introducing the Proposal Distribution: (In this section, the focus is on the upper critical region, i.e., pH values greater than 9.75).

Instead of sampling from the target distribution $f_X(x)$, a new candidate distribution $f_Y(y)$ was selected. Due to computational stability in the weight fraction and easy shift capability, the **Normal distribution** was chosen as the proposal distribution. (Because the Normal distribution can be easily shifted by merely changing its mean, making the task much simpler).

Mean value: 10

Standard Deviation value: 1.5

Calculating Importance Weights: Since we manipulated the original distribution, to prevent Bias and revert the values to their true probability, every generated sample must be corrected by a factor called the importance weight. The weights are calculated as $w(y) = \frac{f_X(y)}{f_Y(y)}$.

Re-estimating the Probability: After generating samples from the new Normal distribution, the probability of the rare event occurrence is calculated via the IS approximation formula:

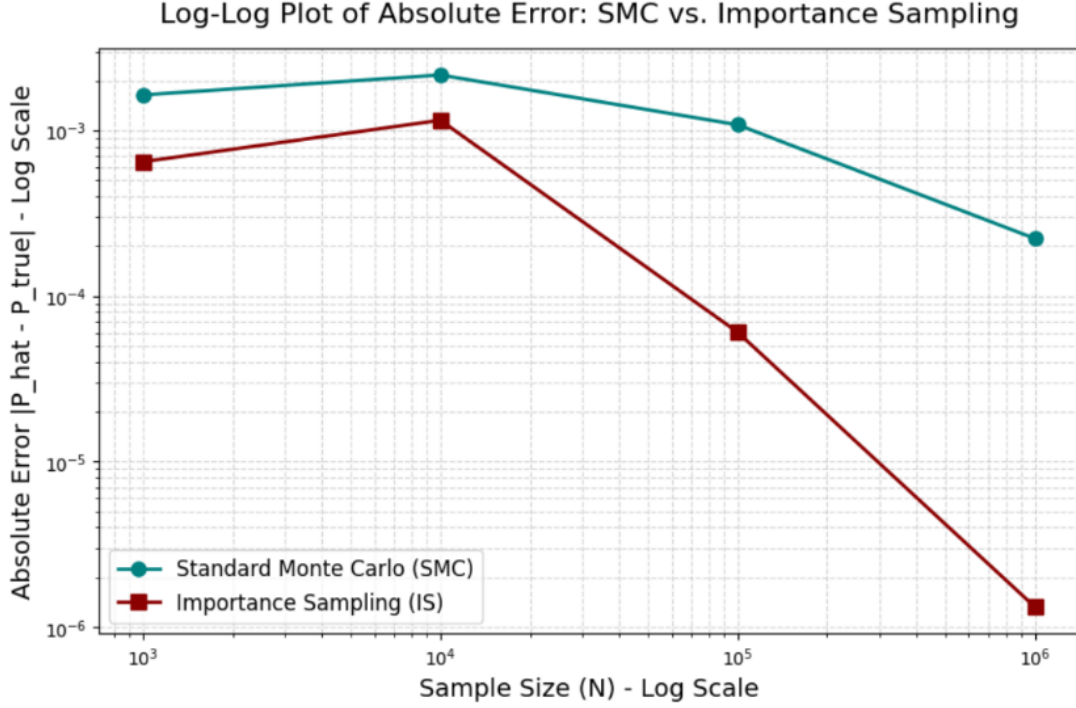
$$\tilde{g} = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{I}(y_i \geq 9.75) \cdot f_X(y_i)}{f_Y(y_i)}$$

The simulation was repeated again for the four sample sizes. The results demonstrated that despite shifting the distribution, the estimator with applied weights converges to the exact same analytical probability of the Gamma distribution. The difference is that now, even in very small sample sizes (e.g., 1000), hundreds of valid critical data points are generated, which elevates the stability and accuracy of the estimation.

$$\tilde{g} = \frac{1}{n} \sum_{i=1}^n \frac{f_X(y_i) \overbrace{g(y_i)}^{\mathbb{I}(y_i \geq 9.75)}}{f_Y(y_i)}$$

6. Evaluation and Results Comparison

--- Importance Sampling Results ---					
N (Sample Size)	IS Estimated Prob	IS Absolute Error	IS Variance (S ² /N)	Rare Events Simulated	
10 ³	0.048009	0.000647	3.841912e-06	576	
10 ⁴	0.046210	0.001152	3.786613e-07	5638	
10 ⁵	0.047422	0.000060	3.913814e-08	56672	
10 ⁶	0.047360	0.000001	3.902903e-09	566632	



The red curve across the entire graph is situated at a significantly lower level than the green curve, demonstrating that by shifting the distribution toward the critical region, the simulation variance is suppressed, and the initial error of the algorithm has been minimized.

After bypassing the initial fluctuations, both lines drop with an almost identical slope. This equal slope proves that both algorithms operate based on the SLLN law; with the difference that the IS method travels this path with less error. Also, looking at the graph, we notice that the error of the SMC method with one million samples is approximately equal to the error of the IS method with only ten thousand samples. This means the IS method achieved the same accuracy consuming only 1% of the time.

7. Final Conclusion

In this project, aiming to model and estimate the probability of rare events in water pH, we compared the efficiency of two statistical simulation techniques. Initially, by moving past raw empirical data and utilizing an analytical approach, the Gamma distribution was fitted as the baseline model. To generate data from this model, the indirect Accept-Reject algorithm with a Laplace envelope was implemented, leading to an acceptance rate of 75%.

Subsequently, evaluating the Standard Monte Carlo method showed that although this approach converges mathematically based on the Strong Law of Large Numbers, due to the extremely small target probability in rare events, it faces the fundamental problem

of high variance and severe waste of time and computation. To overcome this limitation, **Importance Sampling** was employed by shifting the mean of a Normal proposal distribution into the critical region. By applying importance weights and analyzing the Log-Log plot, it became evident that Importance Sampling not only heavily suppresses approximation variance but can achieve the same accuracy as SMC using approximately one percent of computational resources and time.

The results indicate that for critical decision-making in quality engineering and risk management, which require predicting catastrophic failures with extremely low probabilities, utilizing Importance Sampling approaches is a more stable and computationally justifiable solution.

8. Answers to Analytical Questions

- a. **Question 1: How does changing the envelope distribution in the Accept-Reject algorithm affect the acceptance rate? Analyze the relationship between this rate and computational time.**

In the Accept-Reject algorithm, we scale the candidate distribution $g(y)$ by a factor c such that $f(y) \leq c \cdot g(y)$ holds. In this state, the acceptance probability is exactly $p = \frac{1}{c}$.

Changing the distribution $g(y)$ directly changes the optimal value $c = \sup \frac{f(y)}{g(y)}$. If the envelope distribution has a weak match with the distribution $f(y)$, the value of c becomes severely large, and the acceptance rate tends toward zero.

Relationship with computational time: To generate N accepted data points, the computer must run the algorithm loop on average $c \times N$ times. Therefore, as c gets larger, the number of "rejected" samples increases, and computational time increases linearly, making the algorithm inefficient.

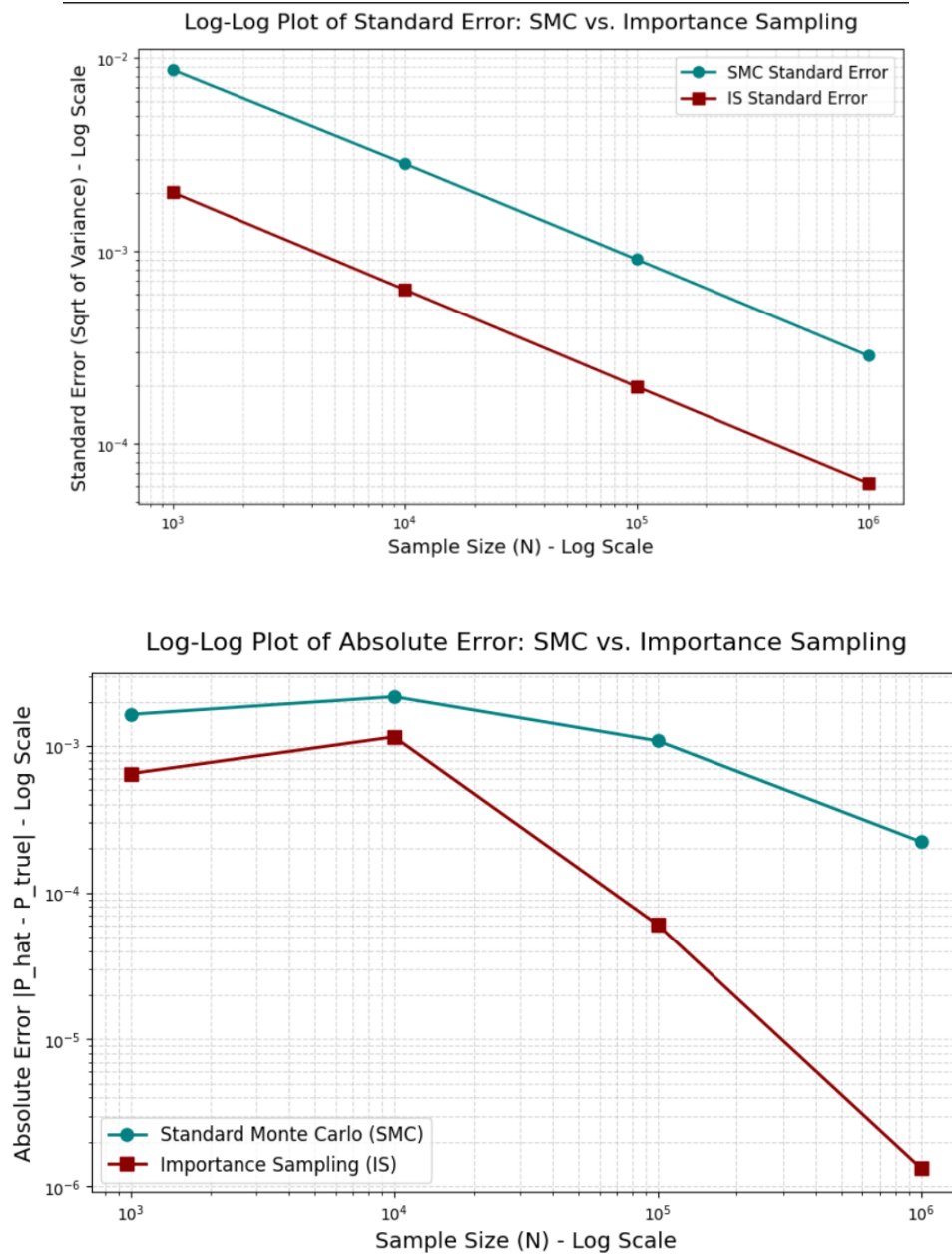
- b. **Question 2: In Importance Sampling, why does choosing a proposal distribution with "heavier tails" than the target distribution reduce the risk of variance instability?**

In Importance Sampling, our estimator depends on "importance weights", defined as $w(y) = \frac{f_X(y)}{f_Y(y)}$.

If f_Y has lighter tails than the target distribution f_X , as y tends to infinity (the tails), the denominator of the fraction tends to zero at a much faster rate than the numerator. This causes the value of the weights to tend to infinity.

Randomly generating a sample with infinite weight makes the approximation entirely unstable. Choosing a distribution with "heavier tails" ensures that the weight fraction always remains bounded and stable.

- c. **Question 3:** By plotting the Log-Log graph for the absolute error versus N , compare the slope of the graph in the two methods of Standard Monte Carlo and Importance Sampling. What does this slope indicate about the convergence rate?



The red curve across the graph sits much lower than the green curve, showing that shifting the distribution minimizes variance. After initial fluctuations, both lines drop with almost identical slopes. This equal slope proves that both algorithms operate on the SLLN law; with the difference that the IS method travels this path with less error. Also, the error of the SMC method with one million samples is approximately equal to the error of the IS method with only ten thousand samples, meaning IS achieved the same accuracy using only 1% of the time.

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- d. **Question 4:** Under what conditions (in terms of distribution shape or threshold value) might the Importance Sampling method have a weaker performance compared to Standard Monte Carlo?

First Case: If the threshold value is close to the population mean (meaning the event is not rare at all), the SMC method is highly efficient on its own. In this state, applying the IS method merely creates additional computational load to calculate PDF functions and weights, offering no advantage.

Second Case: If we place the mean of the proposal distribution (f_Y) too far from the critical region (excessive shift), or if we select a distribution with lighter tails, the problem of most weights approaching zero and one or two weights approaching infinity occurs. In this state, the variance of IS becomes significantly larger than the variance of SMC, rendering the estimation completely invalid.